Adaptive Law-Based Transformation *altx*

Wigner Summer Camp

Data and Compute Intensive Sciences Research Group

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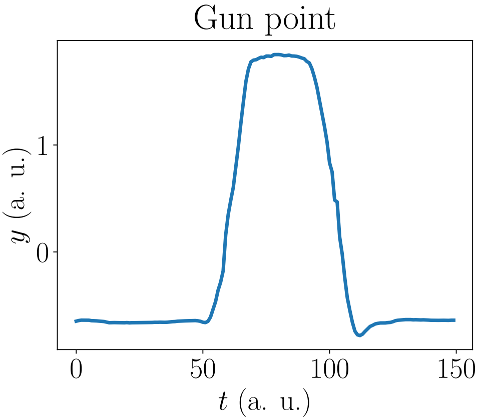
Time series data



Time series is some function of time, for example:

$$x = [1, 2, 5, -1, 3]. \tag{1}$$

Dataset

Consider the following dataset:

- Train instance from class **a**: $a = [1, 1, 2, 3, 5]$.
- Train instance from class **b**: $b = [2, 1, 5, 7, 17]$.
- Test instance from an unknown class: $x = [2, 3, 5, 8, 13]$.

We now apply *altx* to classify the test instance, i.e., determine to which class x belongs.

Goal of *altx*

Evaluate how much the data belongs to one class (**closer to 0, more similar!**)

Data	Class a	Class b
$x_0^{(a)}$	0.2	3
$x_1^{(a)}$	0.03	5
$x_2^{(a)}$	0.1	7
$x_3^{(b)}$	4	0.1
$x_4^{(b)}$	6	0.15
$x_5^{(b)}$	2	0.3
$x^?$	0.00035	0.029

Outline

Step 1: Extracting the Laws

We define parameters $r = 3,\ l = 2,\ k = 1.$ (Their meaning will be discussed later.)

From $a = [1, 1, 2, 3, 5]$ we extract the following segments of length $r = 3$:

$$[1, 1, 2],\ [1, 2, 3],\ [2, 3, 5]. \tag{2}$$

Then we embed them to $l \times l = 2 \times 2$ matrices:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}. \tag{3}$$

From those, the laws[1] vectors which gives close to 0 when multiplying the embedded matrices, are:

$$\begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}, \quad \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}, \quad \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix}. \tag{4}$$

[1] eigenvectors corresponding to the smallest absolute eigenvalues

Step 1: Extracting the Laws (continued)

We combine the results into one matrix and repeat the process for all training instances (separately for each class).
From class **a**, we obtain laws:

$$L_a = \begin{bmatrix} -0.8507 & -0.8507 & -0.8507 \\ 0.5257 & 0.5257 & 0.5257 \end{bmatrix}. \tag{5}$$

From class **b**, we obtain laws:

$$L_b = \begin{bmatrix} -0.9571 & -0.8702 & -0.9085 \\ 0.2898 & 0.4927 & 0.4179 \end{bmatrix}. \tag{6}$$

Step 2: Embedding the Test Instance

The test instance $x = [2, 3, 5, 8, 13]$ is embedded as:

$$E = \begin{bmatrix} 2 & 3 \\ 3 & 5 \\ 5 & 8 \\ 8 & 13 \end{bmatrix}. \tag{7}$$

Step 3: Projection by Laws

Multiplying with class **a** laws:

$$F_a = EL_a = \begin{bmatrix} -0.0344 & -0.0344 & -0.0344 \\ 0.0132 & 0.0132 & 0.0132 \\ -0.0050 & -0.0050 & -0.0050 \\ 0.0019 & 0.0019 & 0.0019 \end{bmatrix}. \tag{8}$$

Multiplying with class **b** laws:

$$F_b = EL_b = \begin{bmatrix} -0.2898 & -0.0727 & -0.1563 \\ -0.2439 & -0.0252 & -0.1091 \\ -0.2615 & -0.0434 & -0.1272 \\ -0.2548 & -0.0365 & -0.1203 \end{bmatrix}. \tag{9}$$

Step 4: Feature Calculation

Using the mean of all squared values:

- Class **a**: mean $= 0.00035$.
- Class **b**: mean $= 0.029$.

Conclusion: The test instance is closer to class **a**.

Exercise

Using *presentation_calculation.ipynb* notebook calculate to which class does

$$x = [3, -1, 5, 3, 13] \tag{10}$$

belongs.

Exercise – solution

Using *presentation_calculation.ipynb* notebook calculate to which class does

$$x = [3, -1, 5, 3, 13] \tag{11}$$

belongs.

Class a score: 0.43, class b score: 0.42 $\Rightarrow$ it belongs to **class b**.

Outline

Dataset

The detailed calculation

The Parameters r, l, k

Meaning of Parameters r, l, k

- r: Length of the original sequence window used for feature extraction.
- l: Embedding size, defines a symmetric $l \times l$ matrix.
- $2l - 1$: Number of equally spaced elements sampled from the window of length r.
- k: Step size (stride) used for shifting the window over the time series.

Example r, l, k values

Consider the following instance:
$x' = [2, 1, 5, 7, 17, 31, 65, 127, 257, 511]$.
Let $r = 5, l = 2$, and $k = 1$. Then $2l - 1 = 3$ and

$$x'_1 = [\boxed{\boxed{2}, 1, \boxed{5}, 7, \boxed{17}}, 31, 65, 127, 257, 511].$$

$$x'_2 = [2, \boxed{\boxed{1}, 5, \boxed{7}, 17, \boxed{31}}, 65, 127, 257, 511].$$

$$x'_2 = [2, 1, \boxed{\boxed{5}, 7, \boxed{17}, 31, \boxed{65}}, 127, 257, 511].$$

$$x'_3 = [2, 1, 5, \boxed{\boxed{7}, 17, \boxed{31}, 65, \boxed{127}}, 257, 511].$$

$$x'_4 = [2, 1, 5, 7, \boxed{\boxed{17}, 31, \boxed{65}, 127, \boxed{257}}, 511].$$

$$x'_5 = [2, 1, 5, 7, 17, \boxed{\boxed{31}, 65, \boxed{127}, 257, \boxed{511}}].$$

Example r, l, k values

Consider the following instance:
$x'' = [2, 1, 5, 7, 17, 31, 65, 127, 257, 511]$.
Let $r = 7, l = 2$, and $k = 2$. Then $2l - 1 = 3$ and

$$x''_1 = [\boxed{\boxed{2}, 1, 5, \boxed{7}, 17, 31, \boxed{65}}, 127, 257, 511].$$

$$x''_2 = [2, 1, \boxed{\boxed{5}, 7, 17, \boxed{31}, 65, 127, \boxed{257}}, 511].$$

Exercise: Parameters r, l, k

Exercise 1. Consider $z = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$ with parameters $r = 5,\ l = 2,\ k = 2$. Mark all windows and the sampled elements inside each window, following the notation of the examples.

Exercise 2. The following windows have been identified on $y = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]$:

$$y_1 = [\boxed{\boxed{10}, 20, 30, \boxed{40}, 50, 60, \boxed{70}}, 80, 90, 100].$$

$$y_2 = [10, 20, 30, \boxed{\boxed{40}, 50, 60, \boxed{70}, 80, 90, \boxed{100}}].$$

Determine the parameters r, l, and k.

Exercise: Parameters r, l, k – Solutions

Exercise 1. With $r = 5,\ l = 2,\ k = 2$: sampling step $= \frac{r-1}{2l-2} = 2$, so $2l - 1 = 3$ elements are sampled per window.

$$z_1 = [\boxed{\boxed{1}, 2, \boxed{3}, 4, \boxed{5}}, 6, 7, 8, 9, 10].$$

$$z_2 = [1, 2, \boxed{\boxed{3}, 4, \boxed{5}, 6, \boxed{7}}, 8, 9, 10].$$

$$z_3 = [1, 2, 3, 4, \boxed{\boxed{5}, 6, \boxed{7}, 8, \boxed{9}}, 10].$$

Exercise 2.

- Each window spans 7 elements $\Rightarrow r = 7$.
- Each window contains $2l - 1 = 3$ sampled elements $\Rightarrow l = 2$.
- Windows start 3 positions apart $\Rightarrow k = 3$.

Summary of ALT Method

- Extract subsequences and embed into symmetric matrices.
- Compute eigenvectors of smallest absolute eigenvalues — these are the laws.
- Multiply embedded test instances with class-specific laws.
- Compute summary statistics (e.g., mean of squared values).
- Classify by choosing the class that minimizes the statistic.